

## AI Toolset



Gen AI

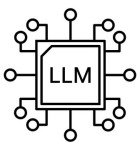
## On Premise Models



## VLLM LLM Pipe



## Task Planning LLM



## Executing LLM



## Orchestration

# 1. Datasets & Harmonization



Gen AI

## Task 1 AI-Guided Data Curation and Harmonization

### 1.1. AI-guided data discovery and organization



Pathology Reports  
+ Clinical Data



Knowledge Base - AJCC



Crawl Data powered by MCP



Data Layer



Scenario Based  
Knowledge Graph

### 1.2. Initial multimodal harmonization and schema-aligned repository construction



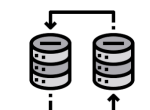
Research  
Keywords  
& Topics



Embeddings



Indexing



Data Fusion



Tagging based on A.



Pre QA/QC



Data Quality Layer

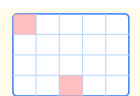
### 1.3. Data quality benchmarking, cataloging, and readiness assessment



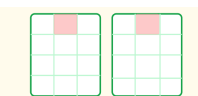
QA/QC



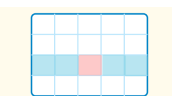
Data Warehouse



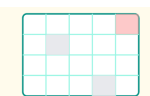
❗ Incomplete records



❗ Cross-source mismatch



❗ Reporting-period mismatch



❗ Low-confidence source

① **Completeness**  
Required fields present  
Record coverage  
Metadata completeness

② **Source /Spatial Consistency**  
Cross-source comparison  
Site/location consistency  
Unit-normalized agreement

③ **Temporal Coherence**  
Time-gap detection  
Reporting-period alignment  
Change-point/outlier screening

④ **Provenance Confidence**  
Source tier weighting  
Expert review status  
Lineage / provenance check

Composite score =  $w_1 \cdot \text{Completeness} + w_2 \cdot \text{SpatialCons.} + w_3 \cdot \text{TempCoh.} + w_4 \cdot \text{Provenance}$

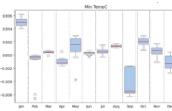
# 2. Analytics AI-enabled workflow

## Task 2 AI-Enhanced EM Analytics

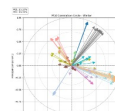
### 2.1. Domain adaptive monitoring and early-warning analytics



Multimodal Data



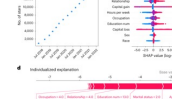
Statistical Test



A. Quality Threshold  
Definition & Parameters



EDA



Variable  
Importance

### 2.2. Cross-domain analytics validation using water/soil data



Breast Cancer



Other Cancer



Pathology Report



Improved  
Data Variety



Human  
Supervision



Statistical  
Analysis



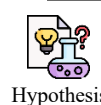
EDA



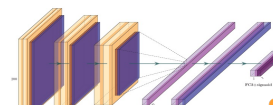
B. Validation

Model Improvement  
GCN, LSTM ....

### 2.3. Model Suite and Providers



Hypothesis



Model Training



OOP Coding



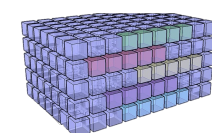
MCP



Weights  
Saving



Provider



Input  
Preparation



Inference



B. Validation

Model Suite

# 3. Disruptive Outcome

## Task 3. Bidirectional HILF

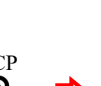
### 3.1. mRAG and human-in-the-loop system



QA/QC'd Data



MCP

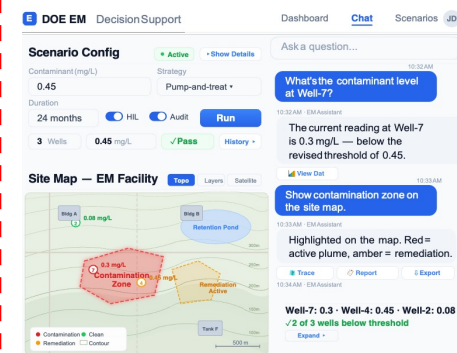


Gen AI

Provider

Model Suite

### 3.2. Bidirectional interface layer



Other Systems



Provider



Gen AI

### 3.3. Feedback-Driven Optimization Or Feedback-Driven Improvement



Generational  
Interface



Knowledge



Feedback



Decision  
Support



Provenance



Human  
Expert

## Rapid Deployment



Docker



Compose

## Scale Up

Horizontal Expansion



Runtime



Kubernetes Node



Proxy

## Sandbox & Development



openstack.

## End User Deployment



AmSC